Few-Shot Object Detection in Remote Sensing Images via Dynamic Adversarial Contrastive-Driven Semantic-Visual Fusion

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***Abstract*—The acquisition of remote sensing images (RSIs) requires expensive equipment, such as satellites or aircraft, along with advanced sensing technologies. Additionally, the complexity of terrains and the inaccessibility of certain regions further exacerbate the difficulty of obtaining RSI data, particularly for novel classes. Therefore, this article focuses on two core challenges in few-shot object detection (FSOD) for RSIs: 1) the difficulty in feature learning caused by insufficient samples of novel classes and 2) large intraclass variability and small interclass differences in novel classes, referring to the feature similarity between different categories and the feature variability within the same category. To address the above two major challenges, we propose an FSOD framework in RSIs that integrates the semantic-visual fusion module (SVFM) with the dynamic adversarial contrastive loss (DACL) function. First, to tackle the issue of insufficient feature representation for novel classes, we introduce a text inference branch that utilizes the CLIP text encoder to generate rich semantic representations. Furthermore, to promote deep integration and synergistic effects between textual information and image features, we design a multilevel cross attention (MLCA) mechanism and a dual-head co-attention (DHCA) guidance feature fusion module to enhance the semantic understanding and visual representation capabilities for novel classes, effectively compensating for the lack of samples. Second, to address interclass similarities and intraclass variabilities, we maximize interclass distances and minimize intraclass feature differences by generating adversarial features and dynamically optimizing feature distributions, thereby enhancing the model's discriminative capability. The experimental results demonstrate that our proposed framework significantly outperforms existing methods across various shot settings on the NWPU VHR-10 and DIOR benchmark datasets for RSIs, showcasing its wide application potential in the field of FSOD in RSIs.**

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***Index Terms*—Contrastive loss function, feature fusion, few-shot object detection (FSOD), remote sensing images (RSIs).**

I. Introduction

THE remote sensing technology acquires ground surface information through aerial and satellite means and has been widely applied in various fields such as disaster monitoring, urban planning, and ecological protection. As one of the key tasks in remote sensing image (RSI) analysis, remote sensing object detection [1], [2], [3], [4] aims to accurately identify specific targets, such as buildings and ships, from images, providing crucial support for decision-making. Early methods relied on handcrafted feature extraction and traditional machine learning algorithms, which struggled to effectively handle complex scenes [5].

In recent years, deep learning-based object detection methods (e.g., Faster R-CNN [6], YOLO [7], and RetinaNet [8]) have significantly improved detection accuracy and robustness by automatically extracting image features, demonstrating exceptional performance on multiple RSI datasets [9], [10], [11]. Despite the advancements in remote sensing object detection driven by deep learning, its heavy reliance on large-scale annotated data limits its applicability. The acquisition of RSIs is influenced by factors such as high equipment costs, long imaging cycles, and complex mission requirements. Additionally, the annotation process demands specialized knowledge, leading to issues of data scarcity and high annotation costs. Particularly, the uneven distribution of novel class targets further exacerbates the challenges of feature learning, resulting in limited performance of traditional detection methods in data-scarce scenarios.

Against this background, few-shot object detection (FSOD) in RSIs has gradually become a research hotspot [12]. Its core idea is to achieve accurate detection of novel category targets with only a small number of annotated samples by improving model design or learning methods. This approach reduces reliance on large-scale datasets, caters to the needs of data-scarce scenarios, and provides a new perspective for addressing the challenges of high data acquisition costs and data scarcity in RSIs. However, research on FSOD in RSIs is still in its infancy. Most existing methods are based on natural images [13], [14], [15], [16], [17] and are difficult to directly apply to RSIs. Compared with natural images, RSIs exhibit

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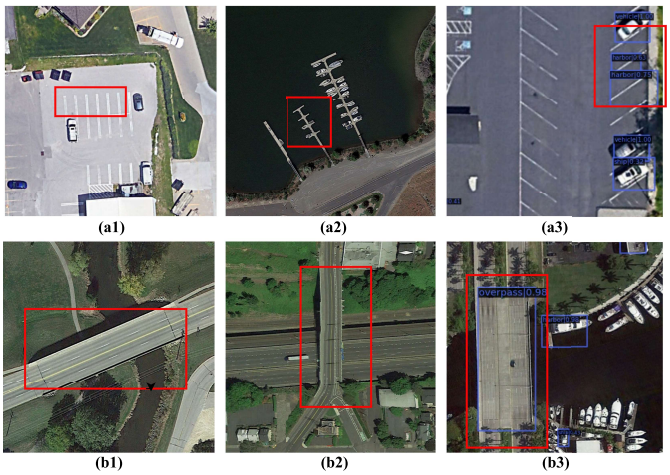


Fig. 1. Visualization of interclass similarity. Image data are sourced from the DIOR dataset. Red bounding boxes indicate the target regions identified by the model during detection. (a1) and (a2) Original images of "Parking lots" and "Harbors," respectively, and (a3) failure case where the model misclassifies a "Parking lot" as a "Harbor." (b1) and (b2) Original images of "Overpass" and "Bridge," respectively, and (b3) failure case where the model misclassifies a "Bridge" as an "Overpass."

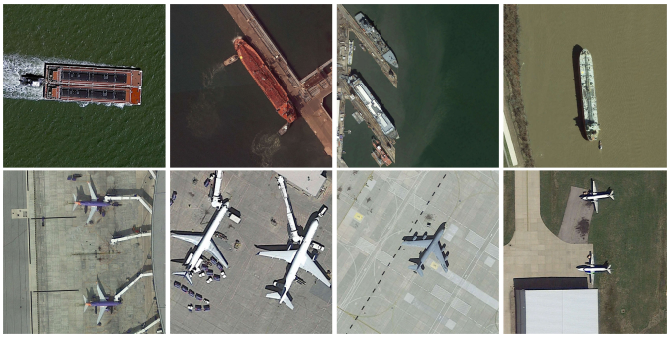


Fig. 2. Illustration of intraclass differences. Images are sourced from the DIOR dataset. The first row displays images of various types of ships, while the second row shows images of different airplane categories. These objects differ significantly in color, shape, and size.

more complex backgrounds, more similar visual features, and sparse target distributions, all of which make the detection task more challenging. Therefore, designing more efficient detection methods that account for the characteristics of RSIs has become an important and pressing issue.

Despite the significant research value of FSOD in RSIs, the unique properties of RSIs pose challenges far beyond those encountered in FSOD tasks for natural images. The primary challenge lies in the extreme scarcity of novel class samples. Compared to base classes, the number of training samples for novel classes is extremely limited and, in some cases, almost nonexistent. This severe data imbalance not only restricts the model's ability to learn features of novel classes, but may also lead to overfitting on base classes during training, thereby neglecting the learning of novel categories. Consequently, the model tends to misclassify novel classes as base classes based on prior knowledge. The second major challenge lies in the interclass similarities and intraclass variabilities within RSIs. Targets from different categories often exhibit similar visual features, such as harbors and parking lots, overpasses and bridges (see Fig. 1). Meanwhile, significant visual differences can also exist within the same category, as shown in the examples of airplanes or ships (see Fig. 2); they exhibit variations in color, model, and size. Additionally, research by Baseline [18] has demonstrated specific instances where detection failures occurred due to the interclass similarities.

To address the challenges of novel class sample scarcity, interclass similarities, and intraclass variability in FSOD for RSIs, this article proposes a framework that integrates the semantic-visual fusion module (SVFM) with the dynamic adversarial contrastive loss (DACL) function. This framework introduces a text inference branch to provide additional semantic information, which is then deeply integrated with image features. Additionally, it designs a novel contrastive loss function, significantly enhancing the model's detection capability and generalization performance for novel classes. Section III details the main components of this approach.

First, based on the traditional object detection framework, this article introduces a text inference branch. By leveraging the pretrained CLIP [19] text encoder, semantic descriptions of novel classes are converted into textual features, which are then deeply integrated with image features to construct a semantically enhanced multimodal feature representation. Specifically, we design a multilevel cross attention (MLCA) mechanism to facilitate deep interaction between textual and image features. Additionally, the dual-head co-attention (DHCA) guidance feature fusion module further enhances the synergistic effects among multimodal features. The incorporation of textual information not only provides additional prior knowledge, but also effectively compensates for the scarcity of novel category samples, significantly improving the detection accuracy for novel classes.

Second, this article proposes a contrastive loss function based on a dynamic adversarial mechanism. This loss function optimizes the model's feature representation capability in few-shot scenarios by maximizing the distance between different class samples and minimizing the distance within the same class samples. Concurrently, we design a dynamic adjustment strategy (DAS) that enables the model to optimize loss function parameters in real time according to feature distribution, thereby achieving optimal performance. Furthermore, we introduce prototype features to cluster similar objects more tightly and design an adversarial feature generation mechanism. This mechanism generates adversarial samples through multistep gradient updates, enhancing the model's robustness in complex backgrounds and sparse target scenarios.

Finally, this article conducts comprehensive experimental validation on two remote sensing datasets: NWPU VHR-10 [20] and DIOR [2]. The experimental results show that compared to existing methods, the proposed approach achieves significant performance improvements across various shot settings. By introducing a joint optimization strategy of text inference and contrastive loss function, this article provides a novel and effective solution for FSOD in RSIs, laying a solid foundation for further research in this field.

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The main contributions of this article can be summarized as follows.

1) This article proposes an SVFM approach. By incorporating additional textual information, the method effectively alleviates class imbalance issues caused by sample scarcity.
2) This article designs a DACL function. By maximizing the distance between different classes and minimizing the distance within the same class, this loss function significantly mitigates the challenge of interclass similarities and intraclass variabilities in RSIs.
3) Extensive experiments were conducted on the NWPU VHR-10 [20] and DIOR [2] datasets, demonstrating that our proposed method shows significant effectiveness in FSOD tasks. We anticipate that these benchmarks will promote further advancements in FSOD within the field of RSIs.

II. Related Work

A. Few-Shot Object Detection

Currently, research on FSOD primarily focuses on two directions: meta-learning [21], [22], [23], [24], [25], [26] and transfer learning [27], [28], [29], [30].

Meta-learning methods aim to endow models with a general learning capability, enabling them to quickly adapt to new tasks or categories. Meta R-CNN [21] introduces a predictor head reconstruction network (PRN) that shares the backbone network with Faster/Mask R-CNN. The PRN receives images containing a few sample objects along with their bounding boxes or masks and infers class attention vectors. On the other hand, FPD [25] generates fine-grained prototypes by distilling the most representative support features and assigns these prototypes to the query feature map based on matching results, thereby modeling detailed feature relationships between the two branches.

Transfer learning achieves effective detection of novel class targets by transferring features from base classes to novel classes. For instance, Gao et al. [27] proposed a method that decouples a standard classifier into two independent modules, where one head focuses on handling clear positive samples, while the other addresses noisy negative samples. Additionally, FSCE [29] introduces a contrastive proposal encoding (CPE) loss, which enhances intraclass compactness and interclass separability at the instance level, effectively alleviating misclassification issues.

However, most of the aforementioned methods are based on natural images and are difficult to directly apply to specialized scenarios such as RSIs. RSIs possess characteristics such as high resolution, complex category distributions, and class imbalance, which impose higher demands on existing FSOD techniques. Therefore, further research into FSOD methods tailored for RSIs holds significant academic value and practical application significance.

B. FSOD in RSIs

FSOD in RSIs, as an emerging research area, has attracted increasing attention in recent years [31], [32], [33], [34], [35], [36]. The rapid development of this field is driven by advancements in deep learning technologies. Specifically, SAE-FSDet [18] proposes a detector tailored for FSOD in RSIs, aiming to prevent significant performance degradation by avoiding label assignment inconsistency and effectively leveraging new instances from the base dataset. ST-FSOD [37] designs a dual-branch region proposal network (RPN) to separate proposal extraction for base and novel objects and introduces a student–teacher mechanism into the RPN and region of interest (RoI) head, treating high confidence but unlabeled targets as pseudolabels. DH-FSDet [38] further enhances the model's flexibility and accuracy by introducing an additional prediction branch specifically for fine-tuning on novel classes.

However, this field still faces numerous challenges, including issues such as sample scarcity, high interclass similarity, and intraclass variability. To address these challenges, researchers have attempted to introduce various strategies, including the use of additional feature information and modifications to the loss function to enhance model performance.

First, regarding the application of additional feature information, researchers have enhanced the model's recognition capability for novel classes by integrating various data sources, such as semantic features, contextual information, and cross-modal features. For instance, TSF-RGR [39] proposes an FSOD method in RSIs based on textual semantic fusion and relational graph reasoning. This approach learns relationships among different categories from commonsense knowledge bases in an end-to-end manner, enabling the detector to perform reasoning across categories. Additionally, TEMO [40] proposes an FSOD method in RSIs based on textual modality knowledge. By incorporating textual descriptions to enhance the semantic information of novel classes, this method designs a textual modality knowledge extractor and a cross-modal fusion module, achieving effective integration of textual and visual features. This significantly alleviates the issue of classification confusion for novel classes. What is more, ViLBERT [41] employs a two-stream architecture to encode images and text separately, achieving multimodal information fusion through cross-attention mechanisms. In contrast, CLIP [19] utilizes contrastive learning to jointly train the ViT image encoder and Transformer-based text encoder. It introduces natural language supervision into large-scale visual pretraining for the first time, enabling zero-shot classification and retrieval without fine-tuning. These methods effectively alleviate the problem of classification confusion caused by limited sample sizes. However, they still rely on simplistic textual features and manually constructed descriptions, failing to fully exploit multilevel semantic interactions. As a result, they introduce higher model subjectivity and manual annotation costs.

Second, regarding the optimization of loss functions, methods such as contrastive loss address the issue of class confusion by maximizing interclass feature distances while minimizing intraclass feature variations. For instance, FSCE [29] incorporates contrastive loss into the object detection framework. By guiding feature learning, it makes features of the same class more compact and enlarges the feature distribution gap between different classes, significantly improving the

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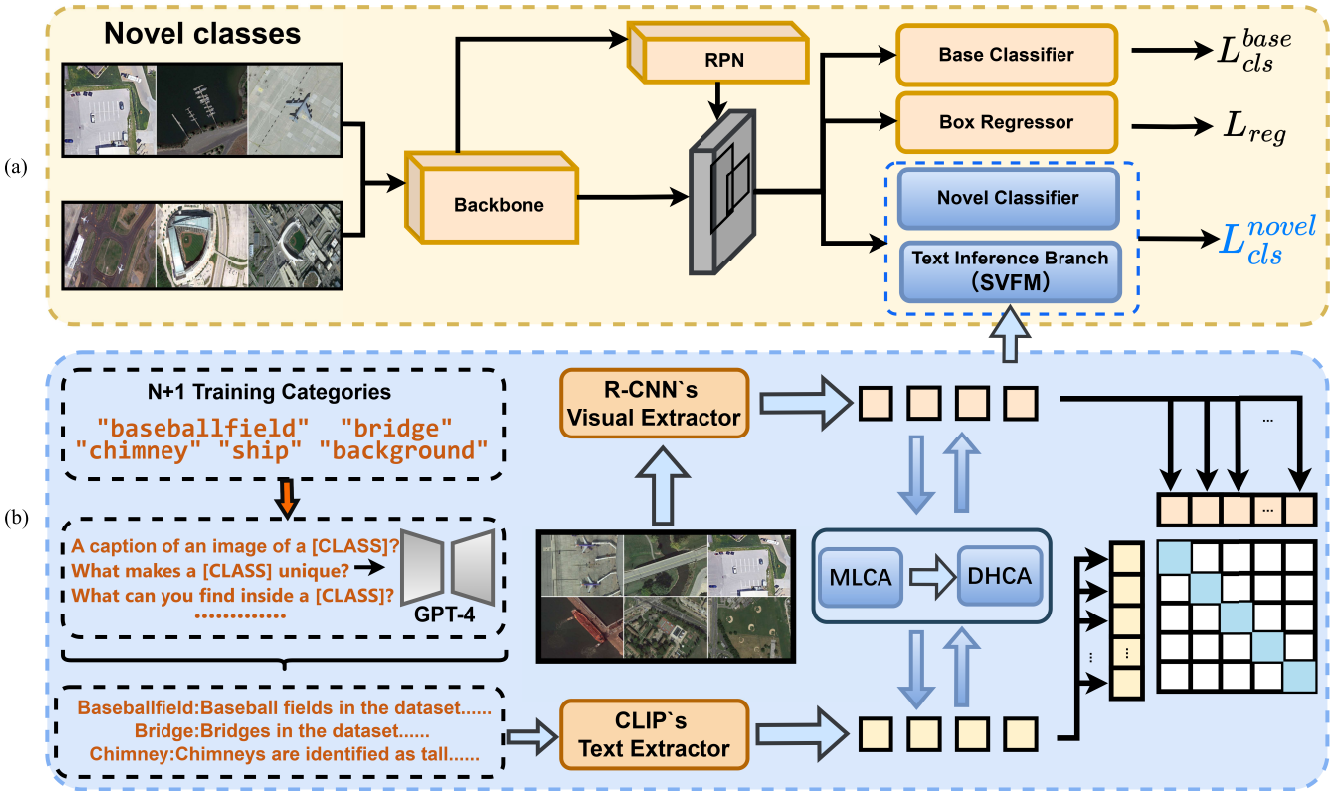


Fig. 3. Illustration of the model architecture. (a) Overall structure of the proposed algorithm. (b) Detailed diagram of the text modality reasoning branch. Please note that the text inference branch corresponds to the flowchart of SVFM.

detection performance for novel classes. Additionally, FADI [42] introduces an additional set-specific margin loss aimed at enhancing interclass separability. Compared to margin-based losses such as ArcFace [43], FADI assigns scaling factors to different margins, with the scaling factor for C_{novel} set higher than that for D_{base}, given the greater challenge posed by novel classes. G-FSDet [44] takes into account the greater intraclass variability and lower interclass separability of geospatial objects, and designs a metric-based discriminative loss to learn a more discriminative classifier during the few-shot fine-tuning stage. TEMO [40] introduces a mask strategy and a separation loss to prevent overfitting and ambiguity in text-model feature. However, these improved methods do not consider dynamically adjusting the loss function to adapt to real-time changes in feature distribution. Moreover, their optimization of the contrastive loss function is insufficient when dealing with the challenges posed by diverse and complex backgrounds. Unlike our method, they do not introduce a dynamic adversarial mechanism to further enhance the model's adaptability across different backgrounds and scenarios. As a result, the generalization and detection capabilities of these models may be limited in complex environments.

To address the aforementioned issues, this study proposes an innovative FSOD framework in RSIs that integrates an SVFM with a DACL function. This framework effectively leverages rich semantic information and interactions between textual and visual features, while optimizing feature distributions to enhance interclass separability and intraclass consistency. As a result, the model achieves significant improvements in both performance and robustness. This integrated strategy effectively tackles the core challenges, bringing new advancements to the field of FSOD in RSIs.

III. Proposed Method

The overall architecture of the proposed method is illustrated in Fig. 3. In this section, we will sequentially introduce the following components: First, Section III-A discusses the data setup for FSOD in RSIs; next, Section III-B provides a detailed description of the SVFM method; and finally, Section III-C elaborates on the design and implementation of the DACL function.

A. Dataset Setup

The FSOD task aims to achieve effective detection of novel classes using a limited number of annotated samples. Compared to traditional object detection methods, the uniqueness of FSOD lies in the design of its data partitioning and training process.

In the FSOD task, the dataset is typically divided into base classes and novel classes. The base classes contain a sufficient number of annotated samples for learning generic feature representations, while the novel classes include only a few samples to simulate data-scarce scenarios. This division ensures that the model can extract rich feature representations from the base classes and effectively transfer them to the novel classes for detection.

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The FSOD process typically consists of two main stages: the base training stage and the fine-tuning stage. In the base training stage, the model is trained solely on base class data to learn generic features and preliminary relationships between categories. During the fine-tuning stage, the model is further optimized using a small number of novel class samples along with randomly sampled base class samples, enhancing its ability to detect objects in the novel classes. In the testing phase, the model is required to detect both base and novel class objects simultaneously to evaluate its generalization performance in few-shot scenarios.

B. SVFM

The scarcity of novel class samples is one of the main challenges in FSOD. Due to the limited number of novel class samples, directly relying on visual features for model training can lead to insufficient feature distribution, thereby affecting the model's classification and detection performance.

To address this issue, we propose an SVFM, the specific architecture of which is illustrated in Fig. 3(b). By introducing a textual reasoning branch, the semantic descriptions of categories are deeply integrated with visual features, significantly enhancing the feature representation capability for rare categories.

Specifically, we design a multimodal feature fusion framework. First, GPT-4 is utilized to generate textual descriptions of categories, which are then encoded into corresponding semantic embeddings using a pretrained CLIP text encoder. Subsequently, an MLCA mechanism and a DHCA guidance fusion module are employed to deeply integrate semantic information with visual features at multiple scales, thereby compensating for the insufficient visual features of novel classes.

1) GPT-4 Prompt Learning: To further enhance the expressiveness of semantic features, this article introduces GPT-4 for category prompt learning to generate richer semantic descriptions. The specific process is as follows. First, GPT-4 generates one or more extended descriptions for each novel class based on predefined prompt templates, covering various aspects such as characteristics and structures of the categories, thereby producing detailed sentences. These descriptions comprehensively capture the diverse semantic information of the categories, effectively addressing the limitations of single textual descriptions. For novel classes with clear semantics and stable structures, a single well-crafted description is sufficient to convey their main characteristics. In contrast, the "background" class typically consists of diverse, unstructured natural scenes, exhibiting high variability. Therefore, we design multiple prompts for this class to more comprehensively cover its potential semantic variations. Let the set of novel classes be $C = \{C_1, C_2, \ldots, C_N\}$, GPT-4 receives a predefined prompt template T and generates an extended semantic description set D_i for each category C_i

$$D_i = \left\{d_{i1}, d_{i2}, \ldots, d_{ij}\right\}, i = 1, 2, \ldots, n. \tag{1}$$

Here, d_{ij} denotes the jth semantic description generated by GPT-4 for category C_i, and m represents the number of semantic descriptions generated for each category. In the experimental setup, the background class is assigned three descriptions ($j = 3$), while each novel class is assigned only one description ($j = 1$).

Subsequently, the CLIP text encoder is utilized to encode the prompt descriptions generated by GPT-4, mapping all semantic information into a unified high-dimensional feature space. During this process, to ensure consistency and generalization of feature dimensions, we average the encoded results of multiple semantic descriptions for each category, thereby generating a unified semantic embedding for the category. This approach not only enhances the representational capability of categories, but also expands the feature space of novel classes through prompt learning, effectively alleviating the issue of sample scarcity.

For each semantic description d_{ij}, the CLIP text encoder f_{clip} is used to map it into the semantic embedding space, yielding the encoded feature vector

$$v_{ij} = f_{\text{clip}}\ \left(d_{ij}\right), v_{ij} \in R^k \tag{2}$$

where k denotes the feature dimension of the CLIP encoder output.

To generate a unified semantic embedding for category C_i, the features of all descriptions are averaged as follows:

$$v_i = \frac{1}{m}\sum_{j=1}^{m} v_{ij}, v_i \in R^k. \tag{3}$$

This approach ensures that the semantic embedding of category C_i covers diverse semantic information while maintaining consistency in feature dimensions.

This prompt learning process is highly automated and highly adaptable, providing the SVFM with high-quality semantic information and significantly enhancing the model's detection performance on novel classes.

In the feature fusion component, this article proposes the MLCA and the DHCA.

2) MLCA Mechanism: Multimodal feature fusion serves as a crucial strategy to address the challenge of insufficient feature representation for novel classes in FSOD. To this end, this article proposes an MLCA. This module enhances the completeness and discriminability of class representations by progressively guiding semantic interactions between visual and textual modalities across layers. Specifically, the visual modality provides local appearance information of objects, such as texture and shape. However, in scenarios where only limited samples of novel classes are available, visual features alone may fail to fully capture the comprehensive characteristics of the categories. To address this limitation, we incorporate semantic priors from the textual modality, including structural descriptions, conceptual relationships, and semantic hierarchies of the categories. These textual features are fused with visual features layer by layer through a cross-modal attention mechanism, enabling the model to form more robust and generalizable joint representations within the semantic space. However, simple fusion methods—such as feature concatenation or weighted summation—may introduce noise due to misalignment between feature spaces, while

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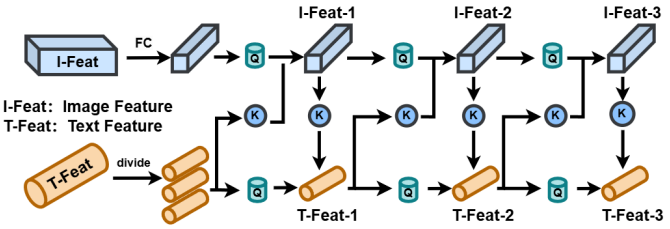


Fig. 4. Illustration of the MLCA. In the figure, I-Feat-1 and T-Feat-1 represent the first-layer image features and first-layer textual features, respectively, while "-2" and "-3" denote the second and third layers. For simplicity, the representation of the V key is omitted in the figure, with the assumption that the input to the V key is identical to that of the K key.

failing to effectively model category-specific properties and semantic relations. In contrast, MLCA achieves deep inter-modal collaboration through a hierarchical dynamic interaction mechanism, which not only suppresses modality-specific noise but also significantly improves the model's ability to distinguish between semantically similar categories.

Specifically, the MLCA achieves deep interaction between modalities through layer-by-layer dynamic updates of textual and visual modality features. The specific architecture is illustrated in Fig. 4. The image features $F_{\text{img}} \in \mathbb{R}^{B\times D}$ output from the ROI, along with the text features $F_{\text{text}} \in \mathbb{R}^{B\times C\times D}$ encoded by CLIP, are utilized, where B denotes the batch size, C represents the number of categories, and D is the feature dimension. To facilitate fusion, the image features are expanded to $F_{\text{img}} \in \mathbb{R}^{B\times 1\times D}$, and the text features are split into multiple subfeatures $F^1_{\text{text}}, F^2_{\text{text}}, \ldots, F^p_{\text{text}}$, each with a shape of $\mathbb{R}^{B\times C\times D}$. Furthermore, to map the image and text modality features into a unified feature space, we generate Key (K), Query (Q), and Value (V) through separate linear transformations

$$\begin{aligned} Q^i_t &= F^i_t W_Q, \\ K^i_t &= F^i_t W_K, t = \text{img}, \text{text}; i = 0, 1, 2, 3, \\ V^i_t &= F^i_t W_V \end{aligned} \tag{4}$$

where W_Q, W_k, and $W_V \in \mathbb{R}^{D\times D_K}$ are the learned parameters, and D_K is the dimensionality of the projected features.

The initial image features Q^0_{img} and textual features K^0_{text}, V^0_{text} are used to compute attention weights, generating the first-layer updated image features

$$F^1_{\text{img}} = \text{Softmax}\left(\frac{Q^0_{\text{img}} K^{0^T}_{\text{text}}}{\sqrt{D_k}}\right) V^0_{\text{text}}. \tag{5}$$

Subsequently, the first-layer updated image features F^1_{img} are used as Key, while the initial textual features F^0_{text} serve as Query and Value, to generate the first layer updated textual features

$$F^1_{\text{text}} = \text{Softmax}\left(\frac{Q^0_{\text{text}} K^{1^T}_{\text{img}}}{\sqrt{D_k}}\right) V^1_{\text{img}}. \tag{6}$$

In subsequent steps, the updated image and textual features are alternately used as inputs for fusion

$$F^i_{\text{img}} = \text{Softmax}\left(\frac{Q^{i-1}_{\text{img}} K^{i-1^T}_{\text{text}}}{\sqrt{D_k}}\right) V^{i-1}_{\text{text}}, i = 1, 2, 3 \tag{7}$$

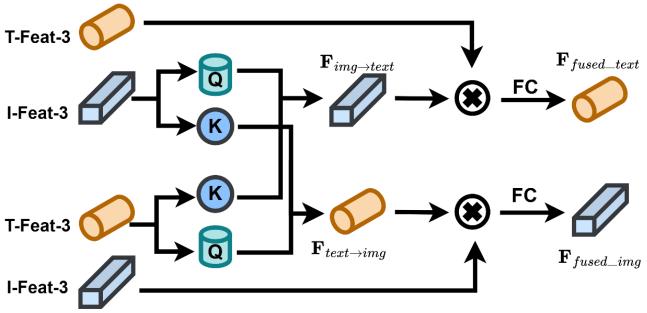


Fig. 5. Architecture of the DHCA. Here, ⊗ denotes the modality concatenation operation, and I-Feat-3 and T-Feat-3 represent the output results from MLCA.

$$F^i_{\text{text}} = \text{Softmax}\left(\frac{Q^{i-1}_{\text{text}} K^{i^T}_{\text{img}}}{\sqrt{D_k}}\right) V^i_{\text{text}}, i = 1, 2, 3. \tag{8}$$

The final output consists of the third-layer image features F^3_{img} and textual features F^3_{text}.

Compared to simple fusion methods, MLCA demonstrates significant advantages in depth and flexibility of feature fusion, making it a key component for enhancing the performance of FSOD.

3) DHCA Guidance Fusion Module: After the feature interaction through the MLCA, the features of the image and text modalities have been preliminarily fused. However, directly using the features output by MLCA still presents two main issues.*Asymmetry in Modal Features:* The image and text features may exhibit misaligned patterns during the fusion process, which affects the discriminative ability of the final fused features for categories. *Insufficient Globality of Fused Features:* While MLCA focuses more on local interactions between modalities, it lacks sufficient information propagation within each modality and global optimization.

To address the aforementioned issues, this article proposes the DHCA to further optimize the image and text modality features. The specific architecture of DHCA is illustrated in Fig. 5. This module employs a bidirectional guided attention mechanism (image-guided text and text-guided image) to deeply integrate the information of the two modalities within the feature space, ensuring alignment between modalities and enhancing category-related information. Additionally, the final fused features are generated through feature concatenation and nonlinear transformation, significantly improving the expressiveness and robustness of the features. With the optimization of DHCA, the final fused features achieve finer cross-modal alignment and capture richer intramodal contextual dependencies, effectively improving the model's detection performance on few-shot and unseen classes.

The DHCA consists of two main steps: bidirectional cross-attention computation and feature concatenation with mapping. The detailed fusion process is as follows.

After processing through MLCA, the image features and text features are denoted as F^3_{img} and F^3_{text}, respectively. Subsequently, the guidance features of the image modality on the text modality, $F_{\text{img}\to\text{text}}$, and the guidance features of the

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text modality on the image modality, $F_{\text{text}\to\text{img}}$, are computed

$$F_{\text{img}\to\text{text}} = \text{Softmax}\left(\frac{Q^3_{\text{img}}K^{3^T}_{\text{text}}}{\sqrt{D_k}}\right)V^3_{\text{text}} \tag{9}$$

$$F_{\text{text}\to\text{img}} = \text{Softmax}\left(\frac{Q^3_{\text{text}}K^{3^T}_{\text{img}}}{\sqrt{D_k}}\right)V^3_{\text{img}}. \tag{10}$$

After completing the bidirectional cross-attention updates, the updated features are concatenated with the original modality features and then mapped back to the original feature space through a fully connected layer

$$F_{\text{fused_img}} = f\left(\text{Concat}\left(F_{\text{img}}, F_{\text{text}\to\text{img}}\right)\right) \tag{11}$$

$$F_{\text{fused_text}} = f\left(\text{Concat}\left(F_{\text{text}}, F_{\text{img}\to\text{text}}\right)\right) \tag{12}$$

where $f(\cdot)$ denotes the fully connected operation, and Concat$(\cdot)$ represents the concatenation operation.

Thus, the combination of DHCA and MLCA forms a powerful semantic-visual feature fusion framework, providing strong support for novel category recognition in FSOD.

The guided image and text features are mapped into the same feature space, and their similarity is computed as

$$\text{Sim_Score} = \cos\left(F_{\text{text}\to\text{img}}, F_{\text{img}\to\text{text}}\right). \tag{13}$$

Each category corresponds to a similarity score, indicating the degree of match between the image features and the textual features of that category. The similarity scores for all categories are concatenated into a tensor $\mathbf{S} \in \mathbb{R}^{B\times C}$, where each element S_{ij} represents the similarity between the ith sample and the jth category, which is used by the subsequent classification head for category prediction.

C. DACL Function

In FSOD tasks, the visual features between categories are often highly similar, especially for novel classes where the class boundaries are relatively ambiguous, leading to confusion between different categories by the model. Additionally, due to the limited number of samples in novel classes, the intraclass feature distribution tends to be unstable, further weakening the model's discriminative ability. These issues significantly constrain the classification accuracy and generalization performance of the model on novel classes. Although some existing contrastive loss functions, such as FSCE [29] and ArcFace [43], attempt to address these challenges by improving the contrastive loss, they typically rely on fixed parameter settings, lacking the necessary adaptability. This limitation hinders their optimization of feature discriminability across different tasks. Furthermore, existing methods fail to enhance the model's generalization capability by expanding the sample distribution, thereby restricting its detection performance on unseen samples.

To address the aforementioned issues, this article proposes the DACL to enhance the discriminative ability between categories while reducing intraclass feature variability. DACL incorporates a DAS, combined with prototype features and an adversarial feature generation module, to expand the feature distribution of novel categories, thereby improving the model's robustness and generalization capability for rare categories.

1) Multistep Gradient Descent for Adversarial Feature: The primary objective of adversarial feature generation is to enhance the representational capability of novel classes by expanding their feature distribution, thereby improving the overall performance of the model. Through an adversarial optimization strategy, adversarial features are generated to be close to the target category center while being distant from other category centers.

Adversarial features are generated through multistep gradient descent and optimized by incorporating dynamic boundaries and adaptive temperature parameters. The specific steps are as follows.

Initially, the adversarial feature is initialized as $f^{(0)}_{\text{adv}} = f$, where f represents the original feature vector. Subsequently, the adversarial feature is updated using a multistep gradient update algorithm

$$f^{(t+1)}_{\text{adv}} = f^{(t)}_{\text{adv}} + \eta \cdot \text{sign}\left(\nabla_{f^{(t)}_{\text{adv}}} L\right) \tag{14}$$

where η denotes the step size. We set η as 0.001 empirically in our experiments. $t = 10$ is the number of gradient updates, and sign$(\cdot)$ represents the symbol function. And L is the contrastive loss computed based on the adversarial feature.

Finally, to control the magnitude of adversarial perturbations, the adversarial feature is projected after each update to ensure that its perturbation does not exceed ϵ

$$\Delta f^{(t+1)}_{\text{adv}} = \text{clamp}\left(f^{(t+1)}_{\text{adv}} - f, -\epsilon, \epsilon\right) \tag{15}$$

where $\Delta f^{(t+1)}_{\text{adv}}$ represents the adversarial perturbation, ϵ denotes the maximum magnitude of the perturbation, and clamp$(x, -\epsilon, \epsilon)$ is an operation that restricts x to the interval $[-\epsilon, \epsilon]$, We set ϵ as 0.01 empirically in our experiments.

Through this approach, it is ensured that the adversarial perturbation does not cause the sample to deviate excessively from its original position, thereby maintaining the sample's validity. The updated adversarial feature is

$$f^{(t+1)}_{\text{adv}} = f + \Delta f^{(t+1)}_{\text{adv}}. \tag{16}$$

After each update, we normalize the adversarial features to ensure that their unit norm remains unchanged

$$f^{(t+1)}_{\text{adv}} = \frac{f^{(t+1)}_{\text{adv}}}{\| f^{(t+1)}_{\text{adv}} \|_2} \tag{17}$$

where $\| \cdot \|_2$ denotes the $L2$ norm, which ensures that the adversarial features maintain a unit norm. This operation helps preserve the consistency and stability of the features during the gradient descent process.

In summary, the total DACL L_{total} is defined as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pro}} + \lambda \cdot \mathcal{L}_{\text{adv}} \tag{18}$$

where L_{adv} is the contrastive loss function based on adversarial features, calculated in the same manner as the contrastive loss for original samples. The weight of the adversarial loss, denoted by λ, is set to 1 by default. L_{pro} is the contrastive loss function based on prototype features, which is described in detail as follows.

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2) Prototype Feature loss Function: This section introduces the design of the contrastive loss L_{pro} based on prototype features. This loss function aims to optimize the feature distribution by generating prototype features, maximizing the distance between categories while minimizing the distance within categories

$$\mathcal{L}_{\text{pro}} = \frac{1}{N}\sum_{i=1}^{N} \max\left(0, S_{\text{neg}} - S_{\text{pos}} + \delta\right) \tag{19}$$

$$S_{\text{pos}} = \frac{\widehat{f_l}^{\top}\mathbf{P}_{y_i}}{\tau}, S_{\text{neg}} = \max_{c\neq y_i}\left(\frac{\widehat{f_l}^{\top}\mathbf{P}_c}{\tau}\right) \tag{20}$$

where P_{y_i} represents the prototype vector corresponding to category y_i, and P_c denotes the prototype vector of other categories c. The term $\max_{c\neq y_i}$ indicates the maximum similarity with the centers of nontarget categories (negative samples). Additionally, τ represents the adaptive temperature parameter, and δ denotes the dynamic boundary.

In this manner, the prototype-based contrastive loss effectively increases the distance between different categories while encouraging samples from the same category to cluster more tightly together, thereby enhancing the model's ability to recognize rare categories.

3) DAS: This section introduces the design of the DAS, which consists of two components. The first component is the dynamic margin adjustment (DMA), which aims to define a dynamically adjusted boundary for the feature distribution of each category pair, ensuring sufficient discriminability between categories. The second component is adaptive temperature scaling (ATS), which dynamically adjusts the sensitivity of feature similarity based on the category distribution, enabling differentiated optimization for novel and base categories.

DMA: The core idea of the dynamic margin is to adjust the discriminative boundary between categories based on the mean similarity of positive samples for each class, thereby more accurately reflecting the density of the feature distribution. Specifically, for categories with a more compact feature distribution, the dynamic margin δ is smaller, whereas for categories with a more dispersed distribution, δ is larger. The calculation formula is as follows:

$$\delta = \lambda \cdot \frac{1}{N_{\text{pos}}}\sum_{k=1}^{N_{\text{pos}}} |S_{\text{pos},k}| \tag{21}$$

where λ is the scaling factor for the dynamic margin, N_{pos} is the number of positive samples, and $|S_{\text{pos},k}|$ represents the absolute value of the similarity of the kth positive sample. We set λ as 1.0 empirically in our experiments.

The dynamic margin can dynamically reflect the distribution state of intraclass features and effectively adjust the boundary range in cases where interclass features are highly similar, thereby reducing confusion. It automatically adapts to the distribution of positive samples, avoiding the limitations of fixed boundary designs.

ATS: The purpose of ATS is to dynamically adjust the sensitivity of similarity calculations based on the standard deviation of the feature distribution. A larger standard deviation typically indicates a more dispersed feature distribution, in which case the sensitivity of similarity calculations should be reduced. Conversely, a smaller standard deviation suggests a more concentrated feature distribution, allowing for increased sensitivity in similarity calculations.

The formula for calculating the adaptive temperature τ is as follows:

$$\tau = \text{std}\left(\left\{\widehat{f_l}\right\}\right) + \epsilon \tag{22}$$

where $\widehat{f_l}$ represents the ith normalized feature vector, std$(\cdot)$ denotes the standard deviation operation, and $\epsilon = 10^{-5}$ is a smoothing term to prevent numerical instability.

This approach allows the model to flexibly adjust the discriminative strength of features based on category distribution, enhancing the ability to distinguish between interclass features. It exhibits greater robustness to the distribution characteristics of novel classes, avoiding the limitations imposed by fixed parameters on model performance.

The DACL significantly improves the model's adaptability to different feature distributions through DMA and ATS. The introduction of adversarial features further enhances the model's recognition capability for complex samples. The experimental results demonstrate that this method shows significant advantages over traditional approaches when handling small-sample datasets with class imbalance and high feature similarity, particularly achieving notable improvements in detection accuracy for rare categories.

IV. Experimental Results

This section sequentially introduces the dataset (Section IV-A), experimental settings (Section IV-B), comparative experiments (Section IV-C), and ablation studies (Section IV-D).

A. Datasets

1) DIOR: The DIOR dataset, developed by Han et al. [2] at XIT University, is a comprehensive benchmark dataset specifically designed for object detection tasks in optical RSIs. The dataset contains 23 463 images with a resolution of 800 × 800 pixels, annotated with 192 472 instances from 20 distinct categories. These categories include infrastructure such as airports and bridges, as well as recreational facilities like stadiums and golf courses. The dataset is divided into 5862 training images, 5863 validation images, and 11 725 testing images, making it a crucial resource for training and evaluating object detection algorithms in remote sensing applications.

2) NWPU VHR-10: The NWPU VHR-10 [45] dataset is a significant resource in the field of remote sensing object detection, comprising ten categories of high-resolution satellite images. It is designed to support the development and evaluation of image analysis algorithms. The dataset is renowned for its diverse imaging conditions and rich object types, making it an ideal choice for testing the robustness of object detection models. The NWPU VHR-10 dataset includes complex scene images capturing various objects such as vehicles, ships, and buildings, providing an ideal testing platform for object detection in real-world environments.

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TABLE I
NOVEL/BASE CLASSES FOR DIFFERENT DATASETS

Data Set	Novel Classes	Base Classes
DIOR [2] Split1	baseballfield, basketballcourt, bridge, chimney, ship	Rest
DIOR [2] Split2	airplane, airport, Expressway-Service-area, harbor, groundtrackfield	Rest
NWPU VHR-10 [45]	airplane, baseball-diamond, tennis-court	Rest
RSOD Split1	aircraft	Rest
RSOD Split2	overpass	Rest

3) RSOD: The RSOD dataset is a high-resolution RSI dataset designed for object detection tasks. It contains approximately 976 optical RSIs with a resolution of 640×640 pixels, covering four typical object categories: aircraft, vehicle, oil tank, and playground. The dataset features accurate annotations, complex scene compositions, and diverse object scales, making it suitable for evaluating and training models dedicated to small-object and multiobject detection in remote sensing imagery. It has been widely adopted in application scenarios such as military reconnaissance and urban planning.

B. Experimental Setting

In the experiments, we follow the baseline settings [13]. For the DIOR [2] dataset, we randomly select 15 categories as base classes and the remaining five categories as novel classes. These novel classes, together with the 15 base classes, form a balanced dataset used for fine-tuning the base model. In the balanced dataset, the number of samples per category k is set to 3, 5, 10, and 20. The novel classes are randomly divided into two groups: the first group includes "baseballfield," "basketballcourt," "bridge," "chimney," and "ship"; the second group includes "airplane," "airport," "Expressway-Service-area," "harbor," and "groundtrackfield," as detailed in Table I.

For the NWPU VHR-10 [45] dataset, we randomly select seven categories as base classes and the remaining three categories as novel classes. These three novel classes, together with the seven base classes, form a balanced dataset used for fine-tuning the base model. In the balanced dataset, the number of samples per category k is also set to 3, 5, 10, and 20. The novel classes are randomly grouped into one set, including "airplane," "baseball-diamond," and "tennis-court," as detailed in Table I.

For the RSOD dataset, which consists of four object classes in total, we follow the dataset partitioning criteria proposed by InfRS [46] by selecting one class as the novel class and the remaining three as base classes. This combination of one novel class and three base classes forms a balanced dataset used for fine-tuning the base model. In this balanced dataset, the number of labeled samples per class, denoted as k, is set to 3, 5, and 10, respectively. The novel class is randomly divided into two groups: the first group includes the category "aircraft," and the second group includes "overpass," as detailed in Table I.

C. Comparative Experiments

1) Results on the DIOR Dataset: To validate the effectiveness of the proposed method, we compared it with several state-of-the-art (SOTA) methods across 3-, 5-, 10-, and 20-shot settings. The experimental results are shown in Table II. Methods such as Meta-RCNN* [21], FsDetView* [22], TFA w/cos* [28], FSCE* [29], ICPE* [47], VFA* [48], and FSOD* [49] were originally designed for natural images. From Table II, it is evident that traditional FSOD methods designed for natural images, such as Meta-RCNN, FsDetView, and TFA w/cos, generally perform worse on RSI detection tasks compared to methods optimized for RSIs. However, some FSOD methods specifically designed for RSIs scored lower than FSCE*, indicating that FSCE* not only performs well on natural images but also shows feasibility in the remote sensing domain.

Furthermore, from Table II, it can be observed that our method achieves the best results across all shot settings in Split 1, strongly demonstrating the superiority and effectiveness of our approach. Compared to the Baseline [18], our method improves performance by 0.65, 1.55, 0.75, and 1.37 points, respectively. In addition, promising results were also achieved in Split 2. Our analysis suggests that the Baseline method, as noted in the original article, tends to misclassify novel classes as similar base classes during detection due to high interclass similarity. In contrast, our method not only provides the model with additional textual feature information, but also optimizes classification by increasing the distance between different classes while reducing the distance within the same class. As a result, our approach is highly effective in addressing misclassification issues.

In addition to demonstrating the effectiveness of our method through evaluation metrics, we also provide visualizations and confusion matrices. Fig. 6 shows the visualization results on the DIOR test set for the ten-shot setting in Split 1. It clearly illustrates the excellent detection performance for both base and novel classes.

2) Results on the NWPU VHR-10 Dataset: To broadly validate the effectiveness of the proposed method, we also conducted comparisons on the NWPU VHR-10 dataset using 3-, 5-, 10-, and 20-shot settings. As shown in Table III, the detection performance of FSCE* is also higher than other methods designed for natural images, indicating that FSCE* possesses strong cross-domain capabilities.

Furthermore, as indicated in Table II, our method consistently achieves the best results across all shot settings, with improvements over the Baseline of 10.5, 11.59, 8.61, and 1.93 points, respectively. This demonstrates the superior effectiveness of our method, especially in small-sample scenarios.

In addition to demonstrating the effectiveness of our method through evaluation metrics, we also provide visualizations of

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TABLE II

ACCURACY COMPARISON ON THE DIOR DATASET, * INDICATES THE ACCURACY RESULTS PROVIDED BY THE BASELINE [18]. THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Split	Method	Year	mAP (DIOR)			
			3-shot	5-shot	10-shot	20-shot
1	Meta-RCNN* [21]	ICCV2019	12.02	13.09	14.07	14.45
	FsDetView* [22]	ECCV2020	13.19	14.29	18.02	18.01
	TFA w/cos* [28]	LCML2020	16.07	15.36	16.45	18.93
	FSCE* [29]	CVPR2021	27.91	28.60	33.05	37.55
	ICPE* [47]	AAAI2023	11.68	12.34	12.95	14.33
	VFA* [48]	AAAI2023	21.94	21.27	23.32	24.28
	FSOD* [49]	CVPR2021	15.94	20.27	24.22	28.16
	P-CNN* [50]	TGRS2021	18.00	22.80	27.60	29.60
	SAGS&TFS [51]	J-STARS2022	**29.30**	31.60	31.60	40.20
	MSOCL [52]	TGRS2022	24.97	27.27	33.37	39.22
	G-FSDet [44]	ISPRS2023	27.57	30.52	**37.46**	39.83
	SAE-FSDet(Baseline)* [18]	TGRS2024	28.80	**32.40**	37.09	**42.46**
	Ours*	-	**29.45**	**33.95**	**37.84**	**43.83**
2	Meta-RCNN* [21]	ICCV2019	8.84	10.88	14.90	16.71
	FsDetView* [22]	ECCV2020	10.83	9.63	13.57	14.76
	TFA w/cos* [28]	LCML2020	6.81	7.53	8.93	11.05
	FSCE* [29]	CVPR2021	13.17	14.07	15.79	20.93
	ICPE* [47]	AAAI2023	10.92	10.56	12.39	13.18
	VFA* [48]	AAAI2023	12.10	12.70	14.72	15.47
	FSOD* [49]	CVPR2021	9.35	9.73	14.84	16.20
	P-CNN* [50]	TGRS2021	**14.50**	14.90	**18.90**	22.80
	SAGS&TFS [51]	J-STARS2022	12.60	15.50	15.50	**23.80**
	MSOCL [52]	TGRS2022	13.31	13.40	15.00	18.15
	G-FSDet [44]	ISPRS2023	14.13	**15.84**	**20.70**	22.69
	SAE-FSDet(Baseline)* [18]	TGRS2024	13.99	15.65	17.41	21.34
	Ours*	-	**14.72**	**16.72**	18.85	**23.02**

TABLE III

ACCURACY COMPARISON ON THE NWPU VHR-10 DATASET. * INDICATES THE ACCURACY RESULTS PROVIDED BY THE BASELINE [14], AND - DENOTES VALUES NOT REPORTED BY THE ORIGINAL AUTHORS. THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Method	Year	mAP (NWPU VHR-10)			
		3-shot	5-shot	10-shot	20-shot
Meta-RCNN* [21]	ICCV2019	20.51	21.77	26.98	28.24
FsDetView* [22]	ECCV2020	24.56	29.55	31.77	32.73
TFA w/cos* [28]	LCML2020	16.17	20.49	21.22	21.57
FSCE* [29]	CVPR2021	41.63	48.80	59.97	79.60
ICPE* [47]	AAAI2023	6.10	9.10	12.00	12.20
VFA* [48]	AAAI2023	13.14	15.08	13.89	20.18
FSOD* [49]	CVPR2021	10.95	15.13	16.23	17.11
P-CNN* [50]	TGRS2021	41.80	49.17	63.29	66.83
TSF-RGR [39]	RS2023	57.00	66.00	77.00	-
G-FSDet [44]	ISPRS2023	49.05	56.10	71.82	75.41
TI-Net [24]	TGRS2023	55.80	63.50	71.80	-
ST-FSOD [37]	TGRS2024	**60.70**	**67.20**	**77.20**	83.30
TEMO [40]	TGRS2024	57.60	66.80	75.10	-
InfRS [46]	TGRS2024	50.00	60.00	74.00	-
FSODM [12]	TGRS2024	32.00	53.00	65.00	-
SAE-FSDet(Baseline)* [18]	TGRS2024	57.96	59.40	71.02	**85.08**
Ours*	-	**68.46**	**70.99**	**79.63**	**87.01**

the results. Fig. 7 shows a comparison between our method and the Baseline on the ten-shot setting. From the figure, it is evident that while the Baseline can detect and localize novel classes under certain conditions, it also suffers from missed detections and misclassifications. For instance, it incorrectly classifies houses as vehicles and fails to detect airplanes, baseball diamonds, and basketball courts.

3) Results on the RSOD: Since the InfRS work did not disclose the specific class partition details of the RSOD dataset, and that this dataset has not been widely used for evaluating FSOD algorithms previously, we could not directly compare our results with those reported in the original article. To address this, based on their setup, we redefined the class partitions and constructed two experimental splits (Splits 1 and 2). Under a unified experimental configuration, we implemented and compared three methods: G-FSDet, SAE-FSDet (Baseline), and ours.

As shown in Table IV, on Split 1, our method achieves mAP scores of 46.64, 55.65, and 64.65 under the three-, five-, and ten-shot settings, respectively, significantly outperforming both G-FSDet (9.09, 9.09, and 15.23) and Baseline (46, 53.41, and 63.23). On Split 2, although the task setting is more challenging, our method still demonstrates strong stability, achieving mAP scores of 16.66, 21.47, and 41.28, respectively. These

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Fig. 6. Visualization of results on the DIOR test set for the Split 1 ten-shot method. Only the bounding boxes with a confidence greater than 0.3 are visualized.

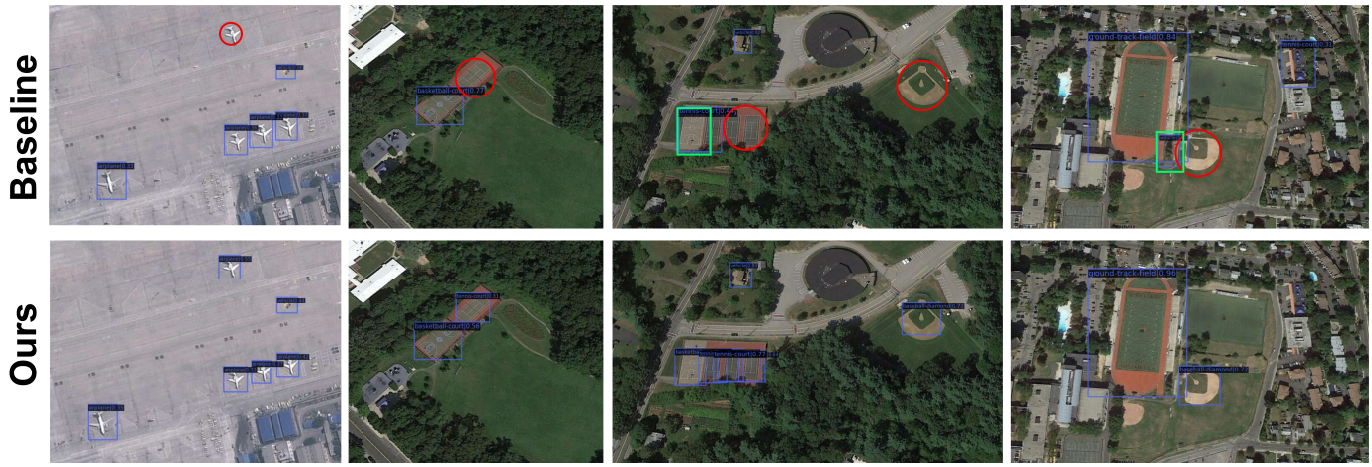


Fig. 7. Visualization of results on the NWPU VHR-10 dataset for the ten-shot setting. The first row shows the visualization results of the Baseline in the ten-shot setting, while the second row presents the results of our proposed method. Red circles indicate missed detections by the Baseline, and green boxes highlight misclassified objects by the Baseline.

results show clear advantages over both Baseline (7, 12.25, and 30.72) and G-FSDet (19.76, 18.09, and 23.56).

These results indicate that, despite the lack of a standardized partition and prior optimization, the proposed method exhibits strong adaptability and robustness on the RSOD dataset. Particularly in few-shot detection scenarios, it demonstrates superior generalization and discrimination capabilities, further validating the effectiveness and generality of our approach.

D. Ablation Studies

In this section, to validate the effectiveness of each module in our method, we conducted detailed ablation studies on the NWPU VHR-10 and DIOR datasets. The results are presented as follows.

1) Ablation Study of Different Modules: We evaluated the SVFM and the DACL by integrating them individually with

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TABLE IV

ACCURACY COMPARISON ON THE RSOD. * INDICATES THE ACCURACY RESULT OBTAINED LOCALLY, AND - DENOTES THE VALUES NOT REPORTED BY THE ORIGINAL AUTHORS. THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Split	Method	Year	mAP(RSOD)		
			3shot	5shot	10shot
1	G-FSDet* [44]	ISPRS2023	9.09	9.09	15.23
	SAE-FSDet(Baseline)* [18]	TGRS2024	**46.00**	**53.41**	**63.23**
	Ours	-	**64.64**	**55.65**	**64.65**
2	G-FSDet* [44]	ISPRS2023	**19.76**	**18.09**	**23.56**
	SAE-FSDet(Baseline)* [18]	TGRS2024	7.00	12.25	30.72
	Ours	-	**16.94**	**20.35**	**37.74**

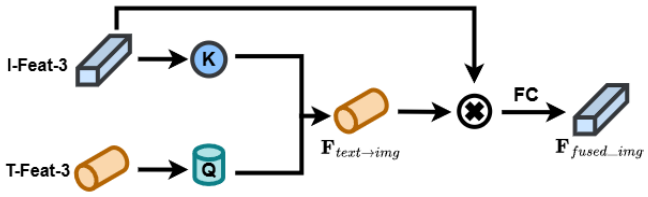


Fig. 8. Structure of SHCA. The distinction from DHCA lies in the fact that this module outputs only the fused image features.

the baseline on both datasets to verify their respective effectiveness. The results are shown in Tables V and VI. From the table, it can be observed that on the DIOR dataset, both SVMF and DACL individually improve performance compared to the Baseline model. Notably, when these two methods are combined (SVFM + DACL), the highest performance gain is achieved, indicating a strong complementary relationship between the two modules, which effectively enhances the overall performance of the model. Similarly, on the NWPU VHR-10 dataset, the same trend is observed. Individual components show improvements over the Baseline, while their combination yields the best results.

These experimental results not only confirm the effectiveness of each individual module, but also highlight the synergistic effects between them. This further demonstrates the advantages of our method in handling complex RSI datasets, especially for FSOD tasks. Additionally, the detailed explanation of the experimental setup and results provides readers with a deeper understanding of how each component impacts model performance, facilitating the future application of these techniques in related research.

2) Ablation Study of SVFM: In this study, we designed ablation experiments specifically for SVFM to evaluate its impact on FSOD tasks in RSIs. As shown in Table VII, the experimental results indicate that each module of SVFM: GPT-4 prompt learning (GPT-4), the MLCA mechanisms, and the DHCA guidance fusion module significantly enhances the model's performance. Furthermore, the integration of these modules leads to even greater improvements.

The significant performance improvement observed in the first and second rows of the table highlights the critical role of GPT-4 in enhancing the model's ability to understand and process limited annotated data, particularly when dealing with complex and variable RSI scenes. Through this innovative learning strategy, our method not only improves the model's recognition accuracy, but also enhances its generalization capability, offering a new technical approach for future RSI analysis tasks.

To further investigate the specific contributions of MLCA and DHCA to model performance, we conducted detailed ablation experiments. The tabulated data in rows 3 and 4 of Table V clearly demonstrate the impact of these two modules on detection performance. From the data, it can be observed that: when using only MLCA, the model performance improves by 3.34%, 5.21%, 3.00%, and 0.28% points compared to the Baseline, indicating that MLCA effectively enhances the model's ability to understand and utilize image features. When using only DHCA, the performance improves by 2.50%, 5.03%, 2.78%, and 0.30% points, demonstrating the significant effectiveness of DHCA in collaborative-guided feature fusion. These results not only validate the effectiveness of these modules as independent modules, but also highlight their substantial contribution to complex RSI analysis tasks when combined. In this way, our research not only enhances the overall performance of the model, but also provides new insights into feature fusion strategies for future studies.

To further optimize the structure of MLCA, we conducted extensive ablation experiments on different numbers of layers within MLCA to investigate their specific impact on model performance and determine the optimal configuration. The experimental results, as shown in Table VIII, illustrate the influence of varying MLCA layer counts on model performance. Specifically, we tested configurations with one, three, and five layers.

One Level: With a single layer, the mAP reaches 61.48%, providing a baseline reference for performance.

Three Levels: Increasing the number of layers to three significantly enhances model performance, achieving the highest mAP of 61.95% among all configurations.

Five Levels: Despite further increasing the layer count to five, the model performance slightly declines, with an mAP of 61.73%, indicating that excessive layers may lead to performance saturation or even a minor degradation.

These results suggest that a three-layer configuration is the most effective for the MLCA, as it enhances the model's feature extraction and processing capabilities while avoiding the burden of over-complexity. Furthermore, through analysis, we find that the lower performance of a single MLCA layer is due to insufficient depth, which limits the model's ability to fully capture the complex relationships between image and text features. On the other hand, while increasing the number of MLCA layers to five enhances model capacity, excessive layers may lead to overfitting, the introduction of feature noise, and performance saturation, thereby reducing the model's generalization capability. Therefore, we recommend adopting the three-layer MLCA configuration in practical applications to achieve optimal performance in RSI object detection tasks. This hierarchical optimization strategy is crucial for designing efficient and accurate deep learning models, particularly when dealing with large-scale data and complex features.

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TABLE V
ABLATION STUDY OF SVFM AND DCAL ON THE DIOR DATASET

Lines	Baseline	SVFM	DACL	mAP (DIOR)			
				3-shot	5-shot	10-shot	20-shot
1	✓			28.80	32.40	37.09	42.46
2	✓	✓		29.25 **(+0.45)**	33.69 **(+1.29)**	37.66 **(+0.57)**	42.99 **(+0.53)**
3	✓		✓	29.27 **(+0.47)**	33.03 **(+0.63)**	37.81 **(+0.72)**	43.81 **(+1.35)**
4	✓	✓	✓	29.45 **(+0.65)**	33.95 **(+1.55)**	37.84 **(+0.75)**	43.83 **(+1.37)**

TABLE VI
ABLATION STUDY OF SVFM AND DACL ON THE NWPU VHR-10 DATASET

Lines	Baseline	SVFM	DACL	mAP (NWPU VHR-10)			
				3-shot	5-shot	10-shot	20-shot
1	✓			57.96	59.40	71.02	85.08
2	✓	✓		61.95 **(+3.99)**	65.36 **(+5.96)**	74.62 **(+3.60)**	85.68 **(+0.60)**
3	✓		✓	67.20 **(+9.24)**	68.26 **(+8.86)**	79.17 **(+8.15)**	86.97 **(+1.29)**
4	✓	✓	✓	68.46 **(+10.50)**	70.99 **(+11.59)**	79.63 **(+8.61)**	87.01 **(+1.93)**

TABLE VII
ABLATION STUDY OF SVFM

Lines	Baseline	GPT-4	MLCA	DHCA	mAP (NWPU VHR-10)			
					3-shot	5-shot	10-shot	20-shot
1	✓				57.96	59.40	71.02	85.08
2	✓	✓			61.67 **(+3.71)**	64.14 **(+4.74)**	73.82 **(+2.80)**	85.15 **(+0.07)**
3	✓		✓		61.30 **(+3.34)**	64.61 **(+5.21)**	74.02 **(+3.00)**	85.36 **(+0.28)**
4	✓			✓	60.46 **(+2.50)**	64.43 **(+5.03)**	73.80 **(+2.78)**	85.38 **(+0.30)**
5	✓	✓	✓	✓	61.95 **(+3.99)**	65.36 **(+5.96)**	74.62 **(+3.60)**	85.68 **(+0.60)**

TABLE VIII
ABLATION STUDY OF MLCA LAYER COUNT

MLCA	mAP (NWPU VHR-10)
	3-shot
1-Level	61.48
3-Level	61.95
5-Level	61.73

TABLE IX
ABLATION STUDY OF DHCA

Method	mAP (NWPU VHR-10)		
	3-shot	10-shot	20-shot
SHCA	61.59	74.53	85.56
DHCA	61.95	74.62	85.68

To further validate the effectiveness of the co-attention guidance fusion module, we conducted ablation experiments comparing dual-head guidance with single-head guidance, aiming to explore the performance improvements achieved by the dual-head structure. The structure of the single-head co-attention (SHCA) guidance fusion module is shown in Fig. 8, and the experimental results are presented in Table IX. As indicated by the data in Table IX, the DHCA outperforms SHCA across all shot settings.

These experimental results demonstrate that dual-head guidance achieves better information complementarity during the feature fusion process, further enhancing the representational power of the features. Compared to SHCA, DHCA significantly improves the model's classification and detection capabilities on limited samples through bidirectional information interaction and multilevel feature integration. This results in stronger robustness and superior performance in RSI's object detection tasks.

3) Ablation Study of DACL: To investigate the specific contributions of the three modules proposed in DACL—DAS, prototype feature loss, and adversarial feature loss—on model performance, we conducted detailed ablation studies for the NWPU VHR-10 dataset. DAS includes the adaptive margin adjustment (DMA) and the adaptive temperature scaling (ATS). The experimental results are presented in Table X.

Table X provides a detailed presentation of the ablation study results for DACL. Rows 2–4 evaluate the model's performance when applying only ATS, only DMA, and both ATS and DMA together, respectively. Rows 5 and 6 individually test the prototype feature loss function and adversarial feature loss function to observe their independent impacts on performance. The final row summarizes the overall effectiveness of our proposed method when all these strategies are combined.

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TABLE X
ABLATION STUDY OF DACL

Lines	Baseline	DAS		Prototype	Adversarial	mAP (NWPU VHR-10)			
		DMA	ATS			3-shot	5-shot	10-shot	20-shot
1	✓					57.96	59.40	71.02	85.08
2	✓			✓	✓	63.86 **(+5.90)**	66.59 **(+7.19)**	77.91 **(+6.89)**	86.45 **(+1.37)**
3	✓		✓	✓	✓	66.14 **(+8.18)**	67.76 **(+8.36)**	77.86 **(+6.84)**	85.99 **(+0.91)**
4	✓	✓		✓	✓	59.01 **(+1.05)**	65.70 **(+6.30)**	77.86 **(+6.84)**	86.56 **(+1.48)**
5	✓	✓	✓	✓		64.76 **(+6.80)**	67.19 **(+7.79)**	77.31 **(+6.29)**	86.82 **(+1.14)**
6	✓	✓	✓		✓	66.33 **(+8.37)**	63.55 **(+4.15)**	77.04 **(+6.02)**	85.16 **(+0.08)**
7	✓	✓	✓	✓	✓	67.20 **(+9.24)**	68.26 **(+8.86)**	79.17 **(+8.15)**	86.97 **(+1.89)**

TABLE XI
ABLATION STUDY OF STEP_SIZE (η)

step_size (η)	mAP (NWPU VHR-10)
	3-shot
0.01	67.08
0.002	66.17
0.001	67.20
0.0005	66.07

Overall, the experimental results validate the effectiveness of each module within DACL and demonstrate that the integration of these modules can fully leverage their respective strengths, thereby significantly enhancing the model's overall performance. This further confirms the advanced nature and effectiveness of DACL in FSOD tasks.

In addition, in this experiment, we conducted an ablation study to investigate the impact of the step size (η) on model performance. The evaluation was performed on the NWPU VHR-10 dataset [45], focusing on its effect on mAP. The experimental results show that when the step size is set to 0.001, the mAP reaches 67.20, achieving the best overall performance. A moderate step size can effectively balance the sufficiency and stability of feature updates during training. When the step size increases to 0.01, the mAP slightly drops to 67.08. An excessively large step size may lead to overly aggressive feature updates, which can undermine model stability. Conversely, when the step size decreases to 0.0005, the mAP further declines to 66.07. In this case, the update magnitude becomes too small, slowing down convergence and leading to insufficient feature learning. In summary, a moderate step size—such as 0.001—can effectively improve model performance. The experimental results are presented in Table XI.

V. CONCLUSION

In this article, we address the challenges of insufficient novel class samples and high interclass similarity misclassification in FSOD tasks for RSIs by proposing a novel detection method. We designed the SVFM and the DACL. The SVFM effectively enhances the feature representation capabilities for novel classes through multimodal feature interactions, while the DACL improves the model's class discrimination and robustness via the dynamic adjustment strategies, the prototype feature loss function, and the adversarial feature loss function.

The experimental results demonstrate that our approach achieves SOAT performance across various shot settings on both the DIOR and NWPU VHR-10 datasets, significantly outperforming multiple SOTA methods.

In future research, we plan to further investigate the impact of missing annotations on detection performance and continue optimizing the proposed detection framework to improve accuracy under scenarios with incomplete labeling. We will explore the adoption of more advanced annotation completion techniques or the design of more robust models, aiming to better address the challenges posed by incomplete or erroneous annotations commonly encountered in RSIs.

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